

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction; Electrode potential definition; Electrochemical Cells

1. Strong electrolytes are those which:
 (A) dissolve readily in water
 (B) conduct electricity
 (C) dissociate into ions even at high concentration
 (D) all of the above

CEC087

Galvanic Cell, Measurements of Electrode Potential, Representation of cell

2. Correct cell reaction for the cell representation $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{MnO}_4^-, \text{Mn}^{2+}, \text{H}^+ | \text{Pt}$ is :
 (A) $\text{MnO}_4^-(\text{aq}) + \text{Fe}^{2+}(\text{aq}) + \text{H}^+(\text{aq}) \longrightarrow \text{Mn}^{2+}(\text{aq}) + \text{Fe}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 (B) $\text{MnO}_4^-(\text{aq}) + 5\text{Fe}^{2+}(\text{aq}) + 8\text{H}^+(\text{aq}) \longrightarrow \text{Mn}^{2+}(\text{aq}) + 5\text{Fe}^{3+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
 (C) $\text{MnO}_4^-(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \longrightarrow \text{Mn}^{2+}(\text{aq}) + \text{Fe}^{3+}(\text{aq})$
 (D) None of the above

CEC110

3. In galvanic cell, the salt bridge is used to
 (A) Complete the circuit
 (B) Reduce the electric resistance in the cell
 (C) Separate cathode from anode
 (D) Carry salts for the chemical reaction

CEC098

Redox electrode

4. In which of the following reaction H_2O_2 acts as reducing agent :
 (A) $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \rightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$ (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 (C) $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$ (D) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$

CEC091

Cell Potential and Its relation with ΔG

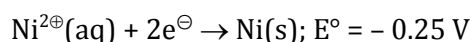
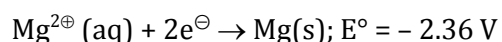
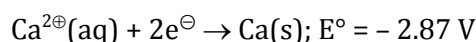
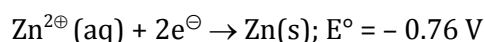
5. Given $E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.36 \text{ V}$, $E^\circ_{\text{Cr}^{3+}/\text{Cr}} = -0.74 \text{ V}$; $E^\circ_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33 \text{ V}$, $E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$.
 Among the following, the strongest reducing agent is :
 (A) Cr (B) Mn^{2+}
 (C) Cr^{3+} (D) Cl^-

CEC101

6. In Cu – Zn cell
 (A) Reduction occurs at the copper cathode
 (B) Oxidation occurs at the copper cathode
 (C) Reduction occurs at the anode
 (D) Chemical energy is converted to light energy

CEC095

7. Consider the following reduction processes :



The reducing power of the metals increases in the order :

- (A) $\text{Ca} < \text{Zn} < \text{Mg} < \text{Ni}$ (B) $\text{Ni} < \text{Zn} < \text{Mg} < \text{Ca}$ (C) $\text{Zn} < \text{Mg} < \text{Ni} < \text{Ca}$ (D) $\text{Ca} < \text{Mg} < \text{Zn} < \text{Ni}$

CEC115

Nernst equation and application of Nernst equation

8. The emf of the cell,

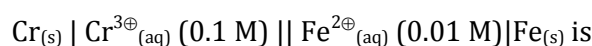


The emf of this cell will be increased by :-

- (A) Increasing the concentration of Cu^{2+} ions (B) Decreasing the concentration of Ti^{+}
(C) Increasing the concentration of both (D) (A) & (B) both

CEC119

9. Given : $E^{\circ}_{\text{Cr}^{3+}/\text{Cr}} = -0.72 \text{ V}$, $E^{\circ}_{\text{Fe}^{2+}/\text{Fe}} = -0.42 \text{ V}$. The potential for the cell



- (A) 0.26 V (B) 0.339 V (C) -0.339 V (D) -0.26 V

CEC123

10. The standard emf of a galvanic cell involving cell reaction with $n = 4$ is found to be 0.295 V at 25°C . The equilibrium constant of the reaction would be,

- (A) 1.0×10^{20} (B) 2.0×10^{11} (C) 4.0×10^{12} (D) 1.0×10^2

CEC125

11. The reduction potential of hydrogen half-cell will be negative if :

- (A) $p(\text{H}_2) = 2 \text{ atm}$ $[\text{H}^{+}] = 1.0 \text{ M}$ (B) $p(\text{H}_2) = 2 \text{ atm}$ and $[\text{H}^{+}] = 2.0 \text{ M}$
(C) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^{+}] = 2.0 \text{ M}$ (D) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^{+}] = 1.0 \text{ M}$

CEC127

12. The cell $\text{Zn} | \text{Zn}^{2+}_{(\text{aq})} (1 \text{ M}) || \text{Cu}^{2+}_{(\text{aq})} (1 \text{ M}) | \text{Cu}$ ($E^{\circ}_{\text{cell}} = 1.10 \text{ V}$) was allowed to be completely discharged at 298 K. The relative concentration of Zn^{2+} to Cu^{2+} , is :

- (A) 9.65×10^4 (B) Antilog (24.08) (C) 37.3 (D) $10^{37.3}$

CEC144

Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction

13. The standard reduction potentials of $\text{Cu}^{2+} | \text{Cu}$ and $\text{Cu}^{2+} | \text{Cu}^{+}$ are 0.337 and 0.153 V respectively. The standard electrode potential of $\text{Cu}^{+} | \text{Cu}$ half cell is :

- (A) 0.184 V (B) 0.827 V (C) 0.521 V (D) 0.490 V

CEC108

14. For the electrochemical cell, $\text{M} | \text{M}^{+} || \text{X}^{-} | \text{X}$, $E^{\circ}_{\text{M}^{+} | \text{M}} = 0.44 \text{ V}$ and $E^{\circ}_{\text{X} | \text{X}^{-}} = 0.33 \text{ V}$. From this data, one can deduce that

- (A) $\text{M} + \text{X} \rightarrow \text{M}^{+} + \text{X}^{-}$ is the spontaneous reaction
(B) $\text{M}^{+} + \text{X}^{-} \rightarrow \text{M} + \text{X}$ is the spontaneous reaction
(C) $E_{\text{cell}} = 0.77 \text{ V}$
(D) $E_{\text{cell}} = -0.77 \text{ V}$

CEC112

15. The standard reduction electrode potentials of four metal cations are -

$$E_{A^{\oplus}/A}^0 = -0.250 \text{ V}, \quad E_{B^{\oplus}/B}^0 = -0.140 \text{ V},$$

$$E_{C^{\oplus}/C}^0 = -0.126 \text{ V}, \quad E_{D^{\oplus}/D}^0 = -0.402 \text{ V}$$

The metal that displaces $A^{\oplus}$ from its aqueous solution is :-

- (A) B (B) C (C) D (D) None of the above

CEC116

16. $E^{\circ}(\text{Ni}^{2\oplus}|\text{Ni}) = -0.25 \text{ volt}$, $E^{\circ}(\text{Au}^{3\oplus}|\text{Au}) = 1.50 \text{ volt}$. The standard emf of the voltaic cell.

$\text{Ni}_{(s)} | \text{Ni}^{2\oplus}_{(aq)} (1.0 \text{ M}) || \text{Au}^{3\oplus}_{(aq)} (1.0 \text{ M}) | \text{Au}_{(s)}$ is :

- (A) 1.25 volt (B) -1.75 volt
(C) 1.75 volt (D) 4.0 volt

CEC142

Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

17. Resistance of 0.2 M solution of an electrolyte is 50 Ω . The specific conductance of the solution is 1.4 S m^{-1} . The resistance of 0.5 M solution of the same electrolyte is 280 Ω . The molar conductivity of 0.5 M solution of the electrolyte in $\text{S m}^2 \text{mol}^{-1}$ is :

- (A) 5×10^3 (B) 5×10^2
(C) 5×10^{-4} (D) 5×10^{-3}

CEC165

18. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100 Ω . The conductivity of this solution is 1.29 Sm^{-1} . Resistance of the same cell when filled with 0.02M of the same solution is 520 Ω . The molar conductivity of 0.02M solution of the electrolyte will be.

- (A) $124 \times 10^{-4} \text{ Sm}^2 \text{mol}^{-1}$ (B) $1240 \times 10^{-4} \text{ Sm}^2 \text{mol}^{-1}$
(C) $1.24 \times 10^4 \text{ Sm}^2 \text{mol}^{-1}$ (D) $12.4 \times 10^{-4} \text{ Sm}^2 \text{mol}^{-1}$

CEC197

19. Electrolytic conduction differs from metallic conduction from the fact that in the former

- (A) The resistance increases with increasing temperature
(B) The resistance decreases with increasing temperature
(C) The resistance remains constant with increasing temperature
(D) The resistance is independent of the length of the conductor

CEC205

Equivalent Conductance; Molar conductance at infinite dilution or limiting molar conductivity

20. The specific conductance of a 0.01 M solution of KCl is 0.0014 $\text{ohm}^{-1} \text{cm}^{-1}$ at 25° C. Its equivalent conductance ($\text{cm}^2 \text{ohm}^{-1} \text{equivalent}^{-1}$) is :-

- (A) 140 (B) 14 (C) 1.4 (D) 0.14

CEC171

21. Which of the following solutions of KCl has the lowest value of equivalent conductance ?

- (A) 1 M (B) 0.1 M (C) 0.01 M (D) 0.001 M

CEC207

Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water

22. At infinite dilution, the equivalent conductance of CH_3COONa , HCl and CH_3COOH are 91, 426 and 391 $\text{mho cm}^2 \text{eq}^{-1}$ respectively at 25°C . The equivalent conductance of NaCl at infinite dilution will be :
 (A) 126 (B) 209 (C) 391 (D) 908

CEC179

23. Molar conductance of 0.1 M acetic acid is $7 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$. If the molar cond. of acetic acid at infinite dilution is $380.8 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$, the value of dissociation constant will be :
 (A) $226 \times 10^{-5} \text{ mol dm}^{-3}$ (B) $1.66 \times 10^{-3} \text{ mol dm}^{-1}$
 (C) $1.66 \times 10^{-2} \text{ mol dm}^{-3}$ (D) $3.442 \times 10^{-5} \text{ mol dm}^{-3}$

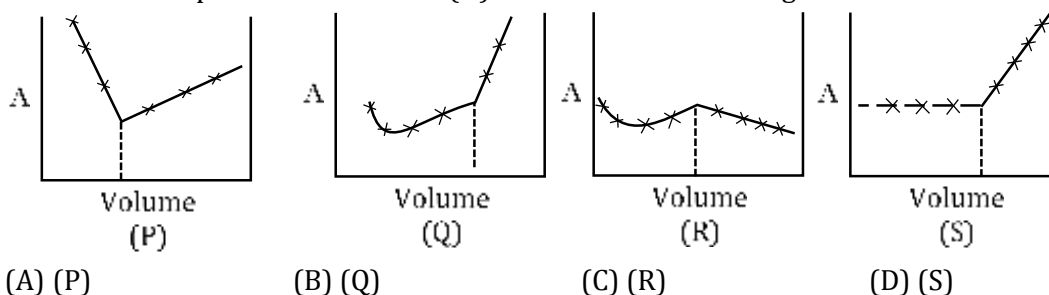
CEC182

24. Equivalent conductance of 0.1 M HA(weak acid) solution is $10 \text{ Scm}^2\text{equivalent}^{-1}$ and that at infinite dilution is $200 \text{ Scm}^2\text{equivalent}^{-1}$ Hence pH of HA solution is
 (A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7

CEC201

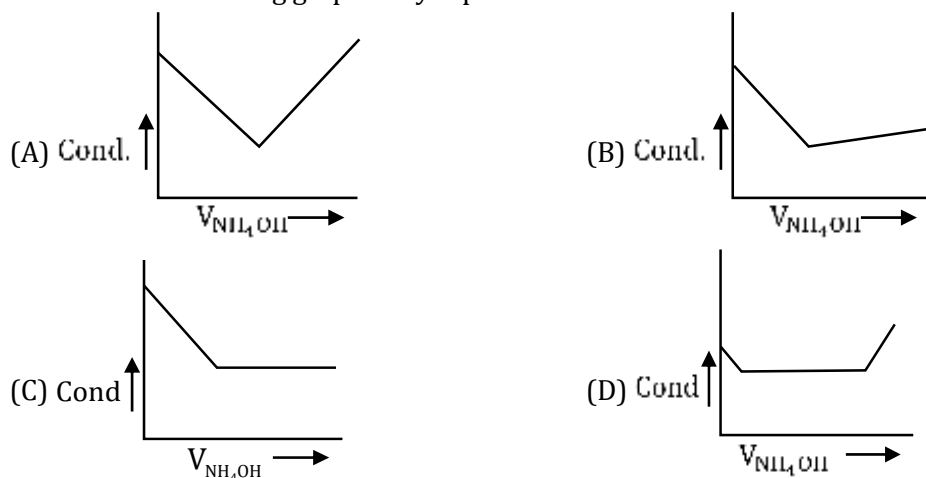
Conductometric titration such as Acid base titrations and precipitation titration; Ionic mobility

25. AgNO_3 was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance (Λ) versus the volume of AgNO_3 is -



CEC185

26. Which of the following graph truly represents the titration of HCl solution against NH_4OH solution ?



CEC188

Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition

27. When an aqueous solution of lithium chloride is electrolysed using graphite electrodes
(A) Cl_2 is liberated at the anode.
(B) Li is deposited at the cathode
(C) as the current flows, pH of the solution remains constant
(D) as the current flows, pH of the solution decreases.

CEC173

Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence

28. In the lead storage battery, the anode reaction is $\text{Pb(s)} + \text{HSO}_4^\ominus + \text{H}_2\text{O} \rightarrow \text{PbSO}_4\text{(s)} + \text{H}_3\text{O}^\oplus + 2\text{e}^\ominus$.
How many grams of Pb will be used up to deliver 1 A for 100 h? (Pb = 208)
(A) 776 g (B) 388 g (C) 194 g (D) 0.1 g

CEC140

29. A solution of sodium sulphate in water is electrolysed using platinum electrodes. The products at cathode and anode are respectively:
(A) H_2 , O_2 (B) O_2 , H_2 (C) O_2 , Na (D) O_2 , SO_2

CEC146

30. The products formed when an aqueous solution of NaBr is electrolyzed in a cell having inert electrodes are :
(A) Na and Br_2 (B) Na and O_2
(C) H_2 , Br_2 and NaOH (D) H_2 and O_2

CEC148

31. How many coulomb of electricity are consumed when 100 mA current is passed through a solution of AgNO_3 for 30 minute during an electrolysis experiment :
(A) 108 (B) 18000 (C) 180 (D) 3000

CEC150

32. When an electric current is passed through acidulated water, 112 ml. of hydrogen gas at STP collects at the cathode in 965 second. The current passed, in ampere is :
(A) 1.0 (B) 0.5 (C) 0.1 (D) 2.0

CEC152

33. A current 0.1 A was passed for 965 second through a solution of $\text{Cu}^\oplus$ solution and 0.03175g of copper was deposited on the cathode. Calculate the current efficiency for the copper deposition. (molecular weight of Cu is 63.5)
(A) 79% (B) 39.5 %
(C) 63.25% (D) 50%

CEC153

34. The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[\text{Ni}(\text{NO}_3)_2]$ and chromium nitrate $[\text{Cr}(\text{NO}_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited is : (at. wt. of Ni = 59, at. wt. of Cr = 52)
(A) 0.1 g (B) 0.17 g (C) 0.3 g (D) 0.6 g

CEC162

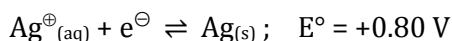
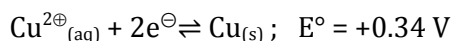
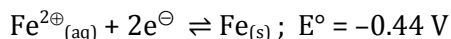
35. The ratio of weights of hydrogen and magnesium deposited by the same amount of electricity from aqueous H_2SO_4 and fused MgSO_4 are :
 (A) 1 : 8 (B) 1 : 12 (C) 1 : 16 (D) None of these
 CEC175
36. One mole of electron passes through each of the solution of AgNO_3 , CuSO_4 and AlCl_3 when Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are
 (A) 1 : 1 : 1 (B) 6 : 3 : 2 (C) 6 : 3 : 1 (D) 1 : 3 : 6
 CEC176
37. 3.17 g., of a substance was deposited by the flow of 0.1 mole of electrons. The equivalent weight of the substance is :
 (A) 3.17 (B) 0.317 (C) 317 (D) 31.7
 CEC213
25. **Primary batteries; Dry cell; Leclanche cell; Mercury cell; Button cell; secondary batteries; Lead storage battery; Nickel Cadmium cell**
38. When a lead storage battery is discharged
 (A) SO_2 is evolved (B) Lead sulphate is consumed
 (C) Lead is formed (D) Sulphuric acid is consumed
 CEC133
39. When lead storage battery is charged
 (A) Lead dioxide dissolves
 (B) Sulphuric acid is regenerated
 (C) The lead electrode becomes coated with lead sulphate
 (D) The amount of sulphuric acid decreases
 CEC136

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Cell Potential and Its relation with ΔG

1. Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (A), (B), (C) and (D).



I. Copper can displace iron from FeSO_4 solution.

II. Iron can displace copper from CuSO_4 solution.

III. Silver can displace copper from CuSO_4 solution.

IV. Iron can displace silver from AgNO_3 solution.

- (A) I and II (B) II and III (C) II and IV (D) I and IV

CEC199

Nernst equation and application of Nernst equation

2. An electrochemical cell is made by combining two half cells (X) $\text{AgNO}_3(0.05) \text{ M} \mid \text{Ag}(\text{s})$ and (Y) $\text{AgNO}_3(0.01) \text{ M} \mid \text{Ag}(\text{s})$. Choose the correct statement

(A) The E_{cell} can be increased if concentration of AgNO_3 in half cell X is increased

(B) The E_{cell} will increase if concentration of AgNO_3 in cell Y is increased

(C) The E_{cell} will increase if concentration of AgNO_3 in cell X is decreased

(D) The E_{cell} will increase if concentration of AgNO_3 in cell Y is decreased

CEC236

Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

3. The specific conductance of saturated solution of silver bromide is $K(\Omega^{-1} \text{ cm}^{-1})$. The limiting ionic conduction of Ag^{+} and Br^{-} ions are a & b respectively. The solubility of AgBr in g lit^{-1} is :

(Molar mass of $\text{AgBr} = 188 \text{ g mol}^{-1}$)

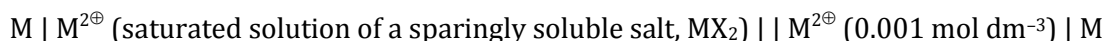
- (A) $K \times \frac{1000}{a-b}$ (B) $\frac{K}{a+b} \times 188$ (C) $\frac{K \times 1000 \times 188}{a+b}$ (D) $\frac{a+b}{K} \times \frac{1000}{188}$

CEC237

COMPREHENSION TYPE QUESTIONS

Paragraph for Question # 4

The electrochemical cell shown below is a concentration cell.



The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes.

The emf of the cell at 298 K is 0.059V.

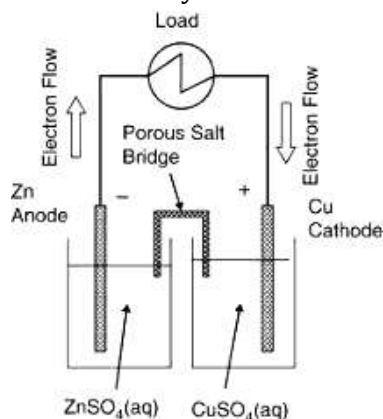
4. The value of ΔG (kJ mol^{-1}) for the given cell is (take $1F = 96500 \text{ C mol}^{-1}$)

- (A) -5.7 (B) 5.7
(C) 11.4 (D) -11.4

CEC211

Paragraph for Question 5 to 6

The most popular electrochemical cell, Daniel cell was originally developed by the English chemist John F. Daniel. The general assembly of the cell is as given below



An undergraduate student setup a Daniel cell using 100 cm³ of 0.100 M CuSO₄ and 100 cm³ 0.100 M ZnSO₄ solution respectively. The two compartments are connected by suitable salt bridge.

(Given = $E^0_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$, $E^0_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.3$)

5. A lab mate of the student asked her for some solid CuCl₂. While she was lining the bottle from a shelf, the lid of the bottle slipped and some amount of CuCl₂ fell in the CuSO₄ compartment at constant volume. She measured the emf of the cell again and found that it had increased by 9 mV. She used this data to calculate the amount of CuCl₂ that had split in the compartment. Calculate the mass of CuCl₂ in grams that had split into the Daniel cell? (Molecular mass of CuCl₂ = 135g)
- (A) 1.35 (B) 13.5 (C) 27 (D) 2.7

CEC238

6. After making this calculation, the student discarded the solution from both the compartments and washed the cell thoroughly. She refilled the compartments with 100 cm³ of the original CuSO₄ and ZnSO₄ solutions. A current of 0.965 A was drawn from the cell for a period of 20 min. What will be the cell potential after 20 min. ?
- (A) 1.091 V (B) 1.109 V (C) 1.118 V (D) 1.082 V

CEC239

Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction

Paragraph for Question 7 to 8

Electrochemical techniques are often used to determine the solubilities of sparingly soluble salts. EMF of a cell is the linear function of logarithm of the concentration of ionic species. This enables us to determine even very small concentration of ions. An electrochemical cell is constructed with one half cell containing Zn(s) rod dipped in 0.2 M Zn(NO₃)₂ (aq.) solution and another half consisting Ag(s) rod immersed in 0.1M AgNO₃ solution. Each solution has a volume of 1L and temperature of 298 K $E^0_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$ $E^0_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$

7. K_{eq} for reaction given below is
- $$2\text{Zn} + 2\text{Ag}^+ (\text{aq.}) \rightleftharpoons 2\text{Ag} + \text{Zn}^{2+} (\text{aq.})$$
- (A) 10⁵² (B) 10⁻⁵² (C) 10²⁶ (D) 10⁻²⁶

CEC240

8. EMF of the above cell reaction becomes 0.981 V and $[K^+] = 0.3 \text{ M}$ upon addition of KCl into AgNO_3 solution. $K_{sp}(\text{AgCl})$ & Cl^- ion concentration in equilibrium with precipitated AgCl respectively are:
- (A) $1 \times 10^{-20} \text{ M}^2$, 0.2 M (B) $2 \times 10^{-29} \text{ M}^2$, 0.2 M
(C) $2 \times 10^{-11} \text{ M}^2$, 0.2 M (D) $2 \times 10^{-11} \text{ M}^2$, 10^{-10} M

CEC241

MATRIX MATCH TYPE QUESTION

Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition

9. Electrolysis of following aqueous solution proceed in electrolytic cell for small period of time.

List-I (Aqueous solution)		List-II (Products)	
(P)	CH_3COOK (Pt electrodes)	(1)	Metal becomes deposited at cathode
(Q)	Dilute NaCl (Pt electrodes)	(2)	H_2 gas at cathode
(R)	CuSO_4 (Pt electrodes)	(3)	H_2 gas at anode
(S)	$\text{Ni}(\text{NO}_3)_2$ (Ni electrodes)	(4)	O_2 gas at anode
		(5)	Concentration of electrolyte remain almost same

	P	Q	R	S
(1)	1	4	2	4
(2)	2	4	2	4
(3)	2	2	4	5
(4)	2	1	4	5

CEC242

Primary batteries; Dry cell; Leclanche cell; Mercury cell; Button cell; secondary batteries; Lead storage battery; Nickel Cadmium cell

10. Statement -1 : The voltage of mercury cell remains constant for long period of time.
Statement -2 : It is because net cell reaction does not involve active species.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CEC138

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Nernst equation and application of Nernst equation

1. $\text{Pt} | \text{H}_2 | \text{H}^+ | | \text{Ag}^+ (0.1\text{M}) | \text{Ag} ; E_{\text{cell}} = 0.86 \text{ volt}$
1 bar

Find pH in anodic half cell

(Given $\frac{2.303RT}{F} = 0.06$; $E_{\text{Ag}^+/\text{Ag}}^0 = 0.8 \text{ volt}$)

CEC129

2. $\text{I}_2(\text{s}) | \text{I}^- (0.1\text{M})$ half-cell is connected to $\text{H}^+_{(\text{aq})} | \text{H}_2(1 \text{ atm}) | \text{Pt}$ half-cell and its half-cell potential is found to be 0.7714 V. If $E_{\text{I}_2/\text{I}^-}^0 = 0.535\text{V}$, the pH of H^+/H_2 half-cell will be :-
 [Give answer as a closest integer]

CEC157

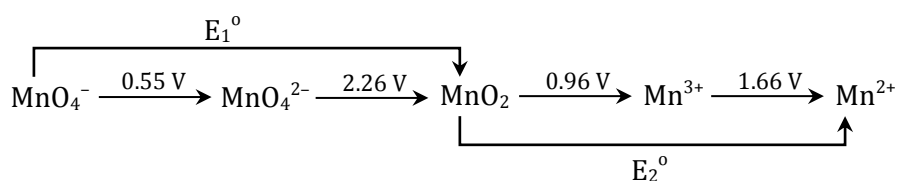
Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction

3. The equilibrium constant for the reaction $\text{Fe} + \text{CuSO}_4 \longrightarrow \text{FeSO}_4 + \text{Cu}$ at 25°C is 2.18×10^y . The value of y is
 Given $E^0(\text{Fe}|\text{Fe}^{2+}) = 0.44\text{V}$, $E^0(\text{Cu}|\text{Cu}^{2+}) = -0.337\text{V}$. [$10^{26.3389} = 2.18 \times 10^{26}$]

CEC159

Latimer diagram

4. From the standard potential in acidic medium as shown in the following latimer diagram, the value of $(E_1^0 + E_2^0)$, in volts, is -



CEC217

Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

5. The molar conductivity of 0.1 M CH_3COOH solution is $4 \text{ S cm}^2 \text{ mol}^{-1}$. What is the resistivity of the solution ?

CEC190

Equivalent Conductance; Molar conductance at infinite dilution or limiting molar conductivity

6. For 0.01N KCl, the resistivity 800 ohm cm. Find equivalent conductivity of KCl in $\text{S cm}^2 \text{ equivalent}^{-1}$?

CEC230

Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water

7. Conductivity of an aqueous solution of 0.1 M HX (a weak mono-protic acid) is $5 \times 10^{-4} \text{ Sm}^{-1}$.

Given : $\Lambda_m^\infty[\text{H}^+] = 0.04 \text{ Sm}^2\text{mol}^{-1}$; $\Lambda_m^\infty[\text{X}^\ominus] = 0.01 \text{ Sm}^2\text{mol}^{-1}$

Find $\text{pK}_a[\text{HX}]$.

CEC202

8. Saturated solution of AgCl at 25°C has specific conductance of $1.12 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$. The $\lambda_\infty(\text{Ag}^\oplus)$ and $\lambda_\infty(\text{Cl}^\ominus)$ are 54 and 58 $\text{ohm}^{-1} \text{ cm}^2/\text{equivalent}$ respectively. The solubility product of AgCl (at 25°C) is 10^{-y} . The value of y is

CEC203

9. Molar conductance of BaCl_2 , H_2SO_4 and HCl at infinite dilutions are $300 \text{ } \Omega^{-1}\text{cm}^2/\text{mole}$, $800 \text{ } \Omega^{-1}\text{cm}^2/\text{mole}$ and $400 \text{ } \Omega^{-1}\text{cm}^2/\text{mole}$ respectively. Equivalent conductance of BaSO_4 at infinite dilution is. (in $\text{S-cm}^2/\text{eq.}$)

CEC232

10. Λ_m° for CH_3COOH based on the data [All data in $\text{S-cm}^2 \text{ mol}^{-1}$].

Electrolyte	CaCl_2	$(\text{CH}_3\text{COO})_2\text{Ca}$	HCl
Λ_m°	1400	800	700

CEC231

Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence

11. After electrolysis of NaCl solution with inert electrodes for a certain period of time. 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu was deposited in the copper voltameter in series with the NaCl electrolytic cell. Calculate the percentage yield of NaOH obtained.

CEC155

12. During electrolysis of NaCl, if 3 moles of NaCl are electrolysed then how much charge in faradays is required if current efficiency is 75% for the process?

CEC227

13. A current of 3.7A is passed for 6hrs. between Ni electrodes in 0.5L of 2M solution of $\text{Ni}(\text{NO}_3)_2$. What will be the molarity of $\text{Ni}(\text{NO}_3)_2$ solution at the end of electrolysis?

CEC228

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Nernst equation and application of Nernst equation

14. At 25°C the value of K_{eq} for the equilibrium $\text{Fe}^{3\oplus} + \text{Ag} \longrightarrow \text{Fe}^{2\oplus} + \text{Ag}^\oplus$ is 0.531 mol/litre. What is the value of $E_{\text{Fe}^{3\oplus}|\text{Fe}^{2\oplus}}^\circ$?

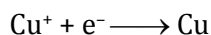
($E_{\text{Ag}^\oplus|\text{Ag}}^\circ = 0.799 \text{ V}$) [$\log (0.531) = -0.27$]

CEC160

15. If for the half cell reactions



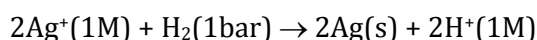
Calculate E° in volts of the half cell reaction



and calculate E° in volts for disproportionation of Cu^+

CEC216

16. Calculate potential in volts of reaction given below :



$$E^\circ_{\text{Ag}^+|\text{Ag}} = 0.80 \text{ V}, E^\circ_{\text{H}^+|\text{H}_2(\text{Pt})} = 0.0 \text{ V}$$

CEC219

17. Equinormal Solutions of two weak acids, HA ($\text{pK}_a = 3$) and HB ($\text{pK}_a = 5$) are each placed in contact with equal pressure of hydrogen electrode at 25°C . When a cell is constructed by interconnecting them through a salt bridge, find the emf of the cell. (in volts)

CEC221

18. In two vessels each containing 500ml water, 0.5m mol of aniline ($K_b = 10^{-9}$) and 50mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf in volts of cell made by connecting them appropriately.

CEC222

Thermodynamics of a cell; Temperature coefficient

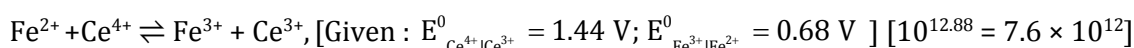
19. For the cell reaction : $\text{Hg}_2\text{Cl}_2(\text{s}) + 2\text{Ag(s)} \rightarrow 2\text{Hg(l)} + 2\text{AgCl(s)}$

temperature coefficient of cell emf is found to be 0.02 VK^{-1} . Find $\Delta_r S^\circ$ for cell reaction in kJ mole^{-1}

CEC224

Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction

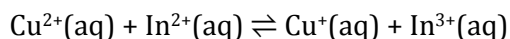
20. If value of equilibrium constant for given reaction is $A \times 10^{12}$ then value of A will be :



CEC223

Latimer diagram

21. Find E° at 298 K for the reaction in volts,



$$\text{Given that } E^\circ_{\text{Cu}^{2+}|\text{Cu}^+} = 0.15 \text{ V}, E^\circ_{\text{In}^{3+}|\text{In}^{2+}} = -0.42 \text{ V}, E^\circ_{\text{In}^{2+}|\text{In}^+} = -0.40 \text{ V}$$

CEC218

Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

22. A solution containing 2.08 g of anhydrous barium chloride in 500 cm^3 of water has a specific conductivity $0.005 \text{ ohm}^{-1}\text{cm}^{-1}$. What are molar and equivalent conductivities of this solution in $\text{mho cm}^2 \text{ mol}^{-1}$ and $\text{mho cm}^2 \text{ equivalent}^{-1}$ respectively.

CEC229

Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water

23. Calculate standard electrode reduction potential in volts when silver wire coated with a layer of silver iodide is placed in solution of KI at 25°C.

$$\text{Given : } E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8 \text{ V ; } K_{\text{sp}}(\text{AgI}) = 10^{-16} ; \frac{2.303 RT}{F} = 0.06$$

CEC220

24. Specific conductance of a saturated solution of AgBr is $8.075 \times 10^{-7} \text{ ohm}^{-1}\text{cm}^{-1}$ at 25°C. Specific conductance of pure water at 25°C is $0.075 \times 10^{-7} \text{ ohm}^{-1} \text{ cm}^{-2}$. Λ_m^∞ for KBr, AgNO₃ and KNO₃ are 140, 130, 110 (S cm² mol⁻¹) respectively. If the solubility of AgBr is ($A \times 10^{-4}$ gm/litre) then value of A will be ? [Atomic mass : Ag = 108 , Br = 80]

CEC233

Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition

25. Calculate mole of electrons lost or gained during electrolysis of
(a) 3.55 gm of Cl⁻ ions (b) 1 gm Cu²⁺ ions
(c) 2.7 gm of Al³⁺ ions
[Atomic mass : Cu = 63.5 , Al = 27, Cl = 35.5]

CEC225

Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence

26. How many faradays of electricity are involved in each of the case
(a) 0.25 mole Al³⁺ is converted to Al.
(b) 27.6 gm of SO₃ is converted to SO₃²⁻
(c) The Cu²⁺ in 1100 ml of 0.5 M Cu²⁺ is converted to Cu.

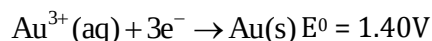
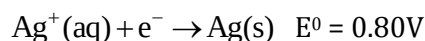
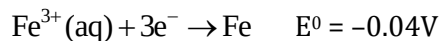
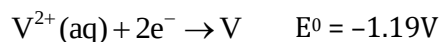
CEC226

EXERCISE – JEE (Main) PYQs

1. The standard electrode potential of M^+/M in aqueous solution does not depend on
 (1) Ionisation of a solid metal atom (2) Sublimation of a solid metal
 (3) Ionisation of a gaseous metal atom (4) Hydration of a gaseous metal ion
[JEE (Main) 2023]
CEC243
2. The reaction

$$\frac{1}{2} \text{H}_2(\text{g}) + \text{AgCl}(\text{s}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{Ag}(\text{s})$$
 occurs in which of the given galvanic cell.
 (1) $\text{Pt}|\text{H}_2(\text{g})|\text{KCl}(\text{sol}^n)|\text{AgCl}(\text{s})|\text{Ag}$ (2) $\text{Pt}|\text{H}_2(\text{g})|\text{HCl}(\text{sol}^n)|\text{AgCl}(\text{s})|\text{Ag}$
 (3) $\text{Ag}|\text{AgCl}(\text{s})|\text{KCl}(\text{sol}^n)|\text{AgCl}(\text{s})|\text{Ag}$ (4) $\text{Pt}|\text{H}_2(\text{g})|\text{HCl}(\text{sol}^n)|\text{AgNO}_3(\text{sol}^n)|\text{Ag}$
[JEE (Main) 2023]
CEC244
3. Standard electrode potentials for a few half cells are mentioned below:
 $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{V}, E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$
 $E_{\text{Ag}^+/\text{Ag}}^0 = 0.80\text{V}, E_{\text{Mg}^{2+}/\text{Mg}}^0 = -2.37\text{V}$
 Which one of the following cells gives the most negative value of ΔG° ?
 (1) $\text{Zn}|\text{Zn}^{2+}(1\text{M})||\text{Ag}^+(1\text{M})|\text{Ag}$ (2) $\text{Zn}|\text{Zn}^{2+}(1\text{M})||\text{Mg}^{2+}(1\text{M})|\text{Mg}$
 (3) $\text{Ag}|\text{Ag}^+(1\text{M})||\text{Mg}^{2+}(1\text{M})|\text{Mg}$ (4) $\text{Cu}|\text{Cu}^{2+}(1\text{M})||\text{Ag}^+(1\text{M})|\text{Ag}$
[JEE (Main) 2025]
CEC293
4. Based on the data given below:
 $E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}}^0 = 1.33\text{V}$ $E_{\text{Cl}_2/\text{Cl}^{(-)}}^0 = 1.36\text{V}$
 $E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.51\text{V}$ $E_{\text{Cr}^{3+}/\text{Cr}}^0 = -0.74\text{V}$
 the strongest reducing agent is :
 (1) Mn^{2+} (2) Cr
 (3) MnO_4^- (4) Cl^-
[JEE (Main) 2025]
CEC294
5. The standard reduction potentials at 298 K for the following half cells are given below :
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}, E^\circ = 1.33\text{V}$
 $\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe} \quad E^\circ = -0.04\text{V}$
 $\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni} \quad E^\circ = -0.25\text{V}$
 $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag} \quad E^\circ = 0.80\text{V}$
 $\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au} \quad E^\circ = 1.40\text{V}$
 Consider the given electrochemical reactions,
 The number of metal(s) which will be oxidized by $\text{Cr}_2\text{O}_7^{2-}$, in aqueous solution is _____.
[JEE (Main) 2024]
CEC246

6. The standard reduction potential at 298 K for the following half cells are given below :-



[JEE (Main) 2023]

The number of metal(s) which will be oxidized by NO_3^- in aqueous solution is _____.

CEC247

7. For a $\text{Mg} | \text{Mg}^{2+}(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag}$ the correct Nernst Equation is :

[JEE (Main) 2025]

$$(1) E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{2F} \ln \frac{[\text{Ag}^+]}{[\text{Mg}^{2+}]} \quad (2) E_{\text{cell}} = E_{\text{cell}}^0 + \frac{RT}{2F} \ln \frac{[\text{Ag}^+]^2}{[\text{Mg}^{2+}]}$$

$$(3) E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{2F} \ln \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]} \quad (4) E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{2F} \ln \frac{[\text{Ag}^+]^2}{[\text{Mg}^{2+}]}$$

CEC295

8. The emf of cell $\text{Tl} \left| \text{Tl}^+ \right|_{(0.001\text{M})} \left| \text{Cu}^{2+} \right|_{(0.01\text{M})} \text{Cu}$ is 0.83 V at 298 K. It could be increased by :

- (1) increasing concentration of Tl^+ ions
- (2) increasing concentration of both Tl^+ and Cu^{2+} ions
- (3) decreasing concentration of both Tl^+ and Cu^{2+} ions
- (4) increasing concentration of Cu^{2+} ions

[JEE (Main) 2024]

CEC248

9. The hydrogen electrode is dipped in a solution of $\text{pH} = 3$ at 25°C . The potential of the electrode will be $-\text{_____} \times 10^{-2} \text{V}$.

$$\left(\frac{2.303RT}{F} = 0.059 \text{V} \right)$$

[JEE (Main) 2024]

CEC249

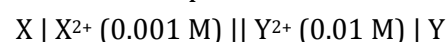
10. At what pH , given half cell $\text{MnO}_4^- (0.1 \text{M}) | \text{Mn}^{2+} (0.001 \text{M})$ will have electrode potential of 1.282 V ? _____ (Nearest Integer)

$$\text{Given } E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.54 \text{V}, \quad \frac{2.303RT}{F} = 0.059\text{V}$$

[JEE (Main) 2023]

CEC250

11. The electrode potential of the following half cell at 298 K.



is $\text{_____} \times 10^{-2} \text{V}$ (Nearest integer).

$$\text{Given : } E_{\text{X}^{2+}/\text{X}}^0 = -2.36\text{V}$$

$$E_{\text{Y}^{2+}/\text{Y}}^0 = +0.36\text{V} \quad \frac{2.303RT}{F} = 0.06\text{V}$$

[JEE (Main) 2023]

CEC252

12. $\text{Pt(s)}|\text{H}_2(\text{g})(1\text{ bar})|\text{H}^+(\text{aq})(1\text{ M})||\text{M}^{3+}(\text{aq}),\text{M}^+(\text{aq})|\text{Pt(s)}$

The E_{cell} for the given cell is 0.1115 V at 298 K when $\frac{[\text{M}^+(\text{aq})]}{[\text{M}^{3+}(\text{aq})]} = 10^a$

The value of a is _____.

Given : $E_{\text{M}^{3+}/\text{M}^+}^0 = 0.2\text{ V}$

$$\frac{2.303 RT}{F} = 0.059\text{ V}$$

[JEE (Main) 2023]

CEC253

13. At 298 K, a 1 litre solution containing 10 mmol of $\text{Cr}_2\text{O}_7^{2-}$ and 100 mmol of Cr^{3+} shows a pH of 3.0.

Given : $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{3+}$; $E^0 = 1.330\text{ V}$ and

$$\frac{2.303 RT}{F} = 0.059\text{ V}$$

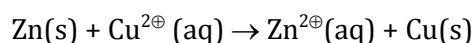
The potential for the half cell reaction is $x \times 10^{-3}\text{ V}$. The value of x is _____.

[JEE (Main) 2023]

CEC049

14. The standard electrode potential E^0 and its temperature coefficient $\left(\frac{dE^0}{dT}\right)$ for a cell are 2V and

$-5 \times 10^{-4}\text{ VK}^{-1}$ at 300 K respectively. The cell reaction is



The standard reaction enthalpy ($\Delta_r H^0$) at 300 K in kJ mol^{-1} is,

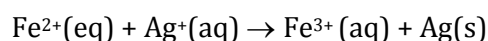
[Use : $R = 8\text{ JK}^{-1}\text{ mol}^{-1}$ and $F = 96,000\text{ Cmol}^{-1}$]

- (1) -412.8 (2) -384.0 (3) 206.4 (4) 192.0

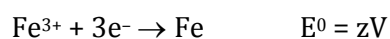
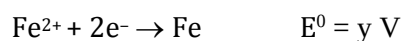
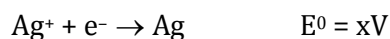
[JEE (Main) 2019]

CEC041

15. For the given cell



The standard cell potential of the above reaction is Given :



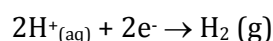
- (1) $x + y - z$ (2) $x + 2y - 3z$ (3) $y - 2x$ (4) $x + 2y$

[JEE (Main) 2025]

CEC296

16. The potential for the given half cell at 298K is

(-)..... $\times 10^{-2}\text{ V}$.



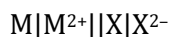
$[\text{H}^+] = 1\text{ M}$, $P_{\text{H}_2} = 2\text{ atm}$

(Given: $2.303 RT/F = 0.06\text{ V}$, $\log 2 = 0.3$)

[JEE (Main) 2024]

CEC254

17. For the electro chemical cell

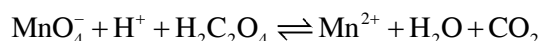


If $E^0_{(M^{2+}/M)} = 0.46 \text{ V}$ and $E^0_{(X/X^{2-})} = 0.34 \text{ V}$. Which of the following is **correct** ?

- (1) $E_{\text{cell}} = -0.80 \text{ V}$
- (2) $M + X \rightarrow M^2 + X^{2-}$ is a spontaneous reaction
- (3) $M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction
- (4) $E_{\text{cell}} = 0.80 \text{ V}$

[JEE (Main) 2024]
CEC255

18. Consider the following redox reaction :



The standard reduction potentials are given as below (E^0_{red})

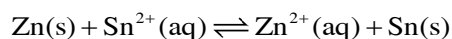
$$E^0_{\text{MnO}_4^-/\text{Mn}^{2+}} = +1.51 \text{ V}$$

$$E^0_{\text{CO}_2/\text{H}_2\text{C}_2\text{O}_4} = -0.49 \text{ V}$$

If the equilibrium constant of the above reaction is given as $K_{\text{eq}} = 10^x$, then the value of $x =$ _____
(nearest integer)

[JEE (Main) 2024]
CEC256

19. The equilibrium constant for the reaction



is 1×10^{20} at 298 K. The magnitude of standard electrode potential of Sn/Sn^{2+} if

$$E^0_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V} \text{ is } _____ \times 10^{-2} \text{ V.}$$

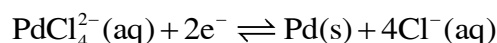
(Nearest integer) Given : $\frac{2.303RT}{F} = 0.059 \text{ V}$

[JEE (Main) 2023]
CEC259

20. The logarithm of equilibrium constant for the reaction $\text{Pd}^{2+} + 4\text{Cl}^- \rightleftharpoons \text{PdCl}_4^{2-}$ is _____.

(Nearest integer)

$$\text{Given: } \frac{2.303RT}{F} = 0.06 \text{ V}$$



$$E^0 = 0.65 \text{ V}$$

[JEE (Main) 2023]
CEC260

21. $\text{FeO}_4^{2-} \xrightarrow{+2.0 \text{ V}} \text{Fe}^{3+} \xrightarrow{0.8 \text{ V}} \text{Fe}^{2+} \xrightarrow{-0.5 \text{ V}} \text{Fe}^0$

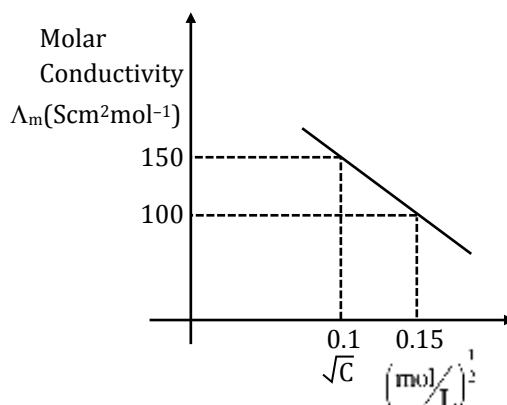
In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E^0_{\text{FeO}_4^{2-}/\text{Fe}^{2+}}$ is

- (1) 1.7 V
- (2) 1.2 V
- (3) 2.1 V
- (4) 1.4 V

CEC297

22. Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.



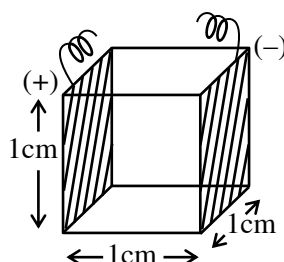
If, for the higher concentration of KCl solution, the resistance of the conductivity cell is 100Ω , then the resistance of the same cell with the dilute solution is 'x' Ω .

The value of x is _____ (Nearest integer)

[JEE (Main) 2025]

CEC298

23. A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will -



- (1) increase sharply
- (2) remain same or can not be measured accurately
- (3) decrease sharply
- (4) depend upon type of electrolyte

[JEE (Main) 2024]

CEC261

24. The values of conductivity of some materials at 298.15 K in Sm^{-1} are 2.1×10^3 , 1.0×10^{-16} , 1.2×10 , 3.91 , 1.5×10^{-2} , 1×10^{-7} , 1.0×10^3 . The number of conductors among the materials is _____.

[JEE (Main) 2024]

CEC262

25. The resistivity of a 0.8 M solution of an electrolyte is $5 \times 10^{-3} \Omega\text{cm}$. Its molar conductivity is _____ $\times 10^4 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$.

(Nearest integer)

[JEE (Main) 2023]

CEC263

26. The number of correct statements from the following is _____. [JEE (Main) 2023]
- (A) Conductivity always decreases with decrease in concentration for both strong and weak electrolytes.
- (B) The number of ions per unit volume that carry current in a solution increases on dilution.
- (C) Molar conductivity increases with decrease in concentration.
- (D) The variation in molar conductivity is different for strong and weak electrolytes.
- (E) For weak electrolytes, the change in molar conductivity with dilution is due to decrease in degree of dissociation.

CEC264

27. The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed? [JEE (Main) 2025]
- (1) A small decrease in molar conductivity is observed at infinite dilution.
- (2) A small increase in molar conductivity is observed at infinite dilution.
- (3) Molar conductivity increases sharply with increase in concentration.
- (4) Molar conductivity decreases sharply with increase in concentration.

CEC299

28. For a strong electrolyte, a plot of molar conductivity against $(\text{concentration})^{1/2}$ is a straight line, with a negative slope, the correct unit for the slope is
- (1) $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$ (2) $\text{S cm}^2 \text{ mol}^{-1} \text{ L}^{1/2}$
- (3) $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}$ (4) $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{-1/2}$

[JEE (Main) 2024]

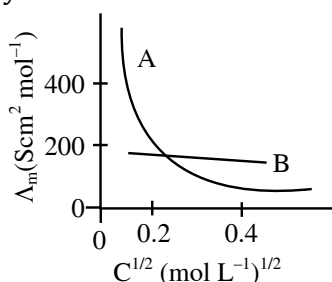
CEC265

29. Molar ionic conductivities of divalent cation and anion are $57 \text{ S cm}^2 \text{ mol}^{-1}$ and $73 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductivity of solution of an electrolyte with the above cation and anion will be :
- (1) $65 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $130 \text{ S cm}^2 \text{ mol}^{-1}$
- (3) $187 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $260 \text{ S cm}^2 \text{ mol}^{-1}$

[JEE (Main) 2024]

CEC266

30. The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are :



- | A | B |
|------------------------|--------------------|
| (1) Weak electrolyte | weak electrolyte |
| (2) Strong electrolyte | strong electrolyte |
| (3) Weak electrolyte | strong electrolyte |
| (4) Strong electrolyte | weak electrolyte |

[JEE (Main) 2024]

CEC267

31. Which out of the following is a correct equation to show change in molar conductivity with respect to concentration for a weak electrolyte, if the symbols carry their usual meaning :

$$\begin{aligned} (1) \quad \Lambda_m^2 C - K_a \Lambda_m^{\circ 2} + K_a \Lambda_m \Lambda_m^{\circ} &= 0 & (2) \quad \Lambda_m - \Lambda_m^{\circ} + AC^{\frac{1}{2}} &= 0 \\ (3) \quad \Lambda_m - \Lambda_m^{\circ} - AC^{\frac{1}{2}} &= 0 & (4) \quad \Lambda_m^2 C + K_a \Lambda_m^{\circ 2} - K_a \Lambda_m \Lambda_m^{\circ} &= 0 \end{aligned}$$

[JEE (Main) 2024]

CEC268

32. At 298 K, the standard reduction potential for $\text{Cu}^{2+} / \text{Cu}$ electrode is 0.34 V.

Given : $K_{sp} \text{Cu(OH)}_2 = 1 \times 10^{-20}$

Take $\frac{2.303RT}{F} = 0.059\text{V}$

The reduction potential at pH = 14 for the above couple is $(-)\text{x} \times 10^{-2}$ V. The value of x is _____.

[JEE (Main) 2023]

CEC269

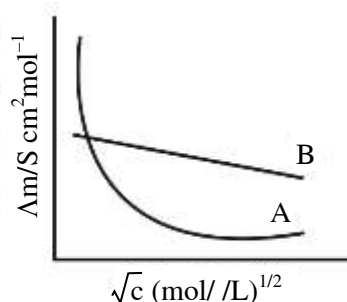
33. The specific conductance of 0.0025 M acetic acid is $5 \times 10^{-5} \text{ S cm}^{-1}$ at a certain temperature. The dissociation constant of acetic acid is _____ $\times 10^{-7}$. (Nearest integer)
Consider limiting molar conductivity of CH_3COOH as $400 \text{ S cm}^2 \text{ mol}^{-1}$

[JEE (Main) 2023]

CEC270

34. Following figure shows dependence of molar conductance of two electrolytes on concentration.

Λ_m° is the limiting molar conductivity.



The number of **incorrect** statement(s) from the following is _____.

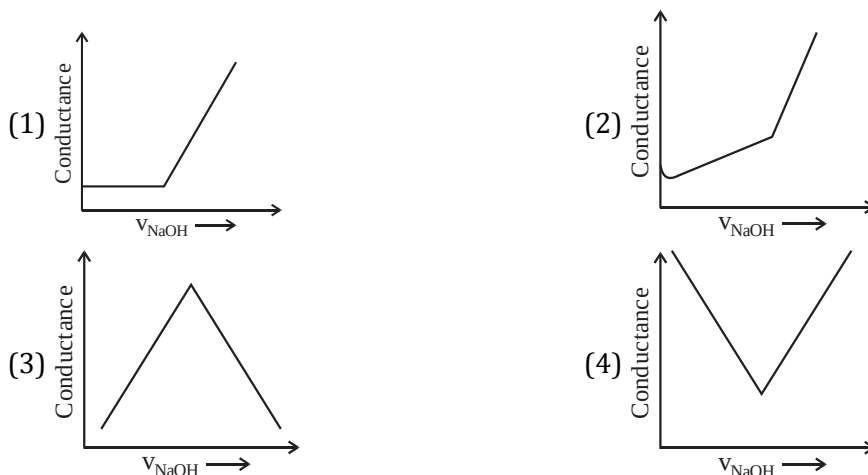
- (A) Λ_m° for electrolyte A is obtained by extrapolation
(B) For electrolyte B, Λ_m vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m°
(C) At infinite dilution, the value of degree of dissociation approach zero for electrolyte B.
(D) Λ_m° for any electrolyte A or B can be calculated using λ° for individual ions.

[JEE (Main) 2023]

CEC272

35. Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.

[JEE (Main) 2023]



CEC050

36. Which of the following electrolyte can be used to obtain $H_2S_2O_8$ by the process of electrolysis?

[JEE (Main) 2025]

- (1) Dilute solution of sodium sulphate
- (2) Dilute solution of sulphuric acid
- (3) Concentrated solution of sulphuric acid
- (4) Acidified dilute solution of sodium sulphate.

CEC300

37. The product, which is not obtained during the electrolysis of brine solution is

[JEE (Main) 2023]

- | | |
|-----------|------------|
| (1) NaOH | (2) Cl_2 |
| (3) H_2 | (4) HCl |

CEC274

38. Which one of the following statements is correct for electrolysis of brine solution?

[JEE (Main) 2023]

- | | |
|---------------------------------|---------------------------------|
| (1) Cl_2 is formed at cathode | (2) O_2 is formed at cathode |
| (3) H_2 is formed at anode | (4) OH^- is formed at cathode |

CEC275

39. A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2A. The amount of the aluminium deposited at the cathode is ____.

[Given : molar mass of aluminium and chlorine are 27 g mol^{-1} and 35.5 g mol^{-1} respectively, Faraday constant = 96500 C mol^{-1}].

[JEE (Main) 2025]

- | | |
|-------------|-------------|
| (1) 1.660 g | (2) 1.007 g |
| (3) 0.336 g | (4) 0.441 g |

CEC301

40. Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12.

The current in Amperes used for the given electrolysis is _____. (Nearest integer).

[JEE (Main) 2025]

CEC302

41. The mass of silver (Molar mass of Ag : 108 g mol^{-1}) displaced by a quantity of electricity which displaces 5600 mL of O_2 at S.T.P. will be ____ g. [JEE (Main) 2024]
CEC276
42. The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is ____ $\times 10^{-4}$ g. (Atomic mass of zinc = 65.4 amu) [JEE (Main) 2024]
CEC277
43. 0.05 cm thick coating of silver is deposited on a plate of 0.05 m^2 area. The number of silver atoms deposited on plate are ____ $\times 10^{23}$. (At mass Ag = 108, $d = 7.9 \text{ g cm}^{-3}$) [JEE (Main) 2024]
CEC279
44. One Faraday of electricity liberates $x \times 10^{-1}$ gram atom of copper from copper sulphate, x is _____. [JEE (Main) 2024]
CEC280
45. A metal surface of 100 cm^2 area has to be coated with nickel layer of thickness 0.001mm. A current of 2A was passed through a solution of $\text{Ni}(\text{NO}_3)_2$ for 'x' seconds to coat the desired layer. The value of x is ____ (Nearest integer) (ρ_{Ni} (density of Nickel) is 10 g mL^{-1} , Molar mass of Nickel is 60 g mol^{-1} $F = 96500 \text{ C mol}^{-1}$) [JEE (Main) 2023]
CEC282
46. The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of PbSO_4 electrolyzed (in g) during the process is : (Molar mass of $\text{PbSO}_4 = 303 \text{ g mol}^{-1}$) [JEE (Main) 2019]
(1) 22.8 (2) 15.2 (3) 7.6 (4) 11.4
CEC044
47. When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :- [JEE (Main) 2018]
(1) 10.9 g (2) 98.1 g
(3) 109.0 g (4) 9.81 g
CEC038
48. The metals that are employed in the battery industries are
A. Fe B. Mn C. Ni
D. Cr E. Cd
Choose the correct answer from the options given below:
(1) B, C and E only (2) A, B, C, D and E
(3) A, B, C and D only (4) B, D and E only [JEE (Main) 2024]
CEC283
49. Identify the factor from the following that does not affect electrolytic conductance of a solution.
(1) The nature of the electrolyte added. (2) The nature of the electrode used.
(3) Concentration of the electrolyte. (4) The nature of solvent used. [JEE (Main) 2024]
CEC284

50. For lead storage battery pick the correct statements

- A. During charging of battery, PbSO_4 on anode is converted into PbO_2
 B. During charging of battery, PbSO_4 on cathode is converted into PbO_2
 C. Lead storage battery, consists of grid of lead packed with PbO_2 as anode
 D. Lead storage battery has $\sim 38\%$ solution of sulphuric acid as an electrolyte

Choose the correct answer from the options given below :

[JEE (Main) 2023]

- (1) B, D only (2) B, C, D only (3) A, B, D only (4) B, C only

CEC285

51. Given below are two statements :

[JEE (Main) 2025]

Statement (I) : Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II) : The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement I and Statement II are false
 (2) Statement I is false but Statement II is true
 (3) Both Statement I and Statement II are true
 (4) Statement I is true but Statement II is false

CEC303

52. Match List I with List II

LIST-I (Cell)		LIST-II (Use/Property/Reaction)	
A.	Leclanche cell	I.	Converts energy of combustion into electrical energy
B.	Ni-Cd cell	II.	Does not involve any ion in solution and is used in hearing aids
C.	Fuel cell	III.	Rechargeable
D.	Mercury cell	IV.	Reaction at anode $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV (2) A-III, B-I, C-IV, D-II
 (3) A-IV, B-III, C-I, D-II (4) A-II, B-III, C-IV, D-I

[JEE (Main) 2024]

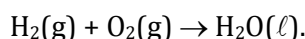
CEC286

EXERCISE - JEE (Advanced) PYQs

1. In a galvanic cell, the salt bridge - [JEE (Advanced) 2014]
 (A) Does not participate chemically in the cell reaction
 (B) Stops the diffusion of ions from one electrode to another
 (C) Is necessary for the occurrence of the cell reaction
 (D) Ensures mixing of the two electrolytic solutions

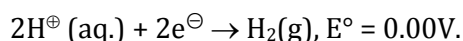
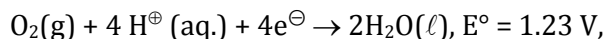
CEC073

2. Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is



The work derived from the cell on the consumption of 1.0×10^{-3} mol of $\text{H}_2(\text{g})$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulated container. What is the change in the temperature (in K) of the ideal gas ?

The standard reduction potentials for the two half-cells are given below.



Use $F = 96500 \text{ C mol}^{-1}$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$

[JEE (Advanced) 2020]

CEC081

3. For the following cell :
 $\text{Zn}(\text{s}) | \text{ZnSO}_4(\text{aq.}) || \text{CuSO}_4(\text{aq.}) | \text{Cu}(\text{s})$
 when the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+} , the expression for ΔG (in J mol^{-1}) is
 [F is Faraday constant, R is gas constant, T is temperature, $E^\circ(\text{cell}) = 1.1 \text{ V}$] [JEE (Advanced) 2017]
 (A) $2.303 RT + 1.1F$ (B) $2.303 RT - 2.2F$ (C) $1.1 F$ (D) $-2.2 F$

CEC078

4. For the electrochemical cell,
 $\text{Mg}(\text{s}) | \text{Mg}^{2+}(\text{aq, 1 M}) || \text{Cu}^{2+}(\text{aq, 1M}) | \text{Cu}(\text{s})$
 the standard emf of the cell is 2.70 V at 300 K. When the concentration of Mg^{2+} is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is ____.

(Given, $\frac{F}{R} = 11500 \text{ KV}^{-1}$, where F is the Faraday constant and R is the gas constant, $\ln(10) = 2.30$)

[JEE (Advanced) 2018]

CEC080

5. For the following electrochemical cell at 298K,
 $\text{Pt}(\text{s}) | \text{H}_2(\text{g, 1bar}) | \text{H}^+(\text{aq, 1M}) || \text{M}^{4+}(\text{aq}), \text{M}^{2+}(\text{aq}) | \text{Pt}(\text{s})$
 $E_{\text{cell}} = 0.092 \text{ V}$ when $\frac{[\text{M}^{2+}(\text{aq.})]}{[\text{M}^{4+}(\text{aq.})]} = 10^x$

Given : $E_{\text{M}^{4+}/\text{M}^{2+}}^0 = 0.151 \text{ V}$; $2.303 \frac{RT}{F} = 0.059 \text{ V}$

The value of x is -

[JEE (Advanced) 2016]

- (A) -2 (B) -1 (C) 1 (D) 2

CEC076

6. Some standard electrode potentials at 298 K are given below:

Pb^{2+}/Pb	-0.13 V
Ni^{2+}/Ni	-0.24 V
Cd^{2+}/Cd	-0.40 V
Fe^{2+}/Fe	-0.44 V

To a solution containing 0.001 M of X^{2+} and 0.1 M of Y^{2+} , the metal rods X and Y are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of X. The correct combination(s) of X and Y, respectively, is (are)

(Given: Gas constant, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, Faraday constant, $F = 96500 \text{ C mol}^{-1}$)

[JEE (Advanced) 2022]

- (A) Cd and Ni (B) Cd and Fe
(C) Ni and Pb (D) Ni and Fe

CEC084

7. Consider an electrochemical cell: $\text{A(s)} | \text{A}^{n+}(\text{aq}, 2\text{M}) || \text{B}^{2n+}(\text{aq}, 1\text{M}) | \text{B(s)}$. The value of ΔH^0 for the cell reaction is twice that of ΔG^0 at 300 K. If the emf of the cell is zero, the ΔS^0 (in $\text{JK}^{-1} \text{ mol}^{-1}$) of the cell reaction per mole of B formed at 300 K is ____.

(Given : $\ln(2) = 0.7$, R (universal gas constant) = $8.3 \text{ J K}^{-1} \text{ mol}^{-1}$. H , S and G are enthalpy, entropy and Gibbs energy, respectively.)

[JEE (Advanced) 2018]

CEC079

8. The reduction potential (E^0 , in V) of $\text{MnO}_4^- (\text{aq})/\text{Mn(s)}$ is ____.

[JEE (Advanced) 2022]

[Given : $E^0_{(\text{MnO}_4^- (\text{aq})/\text{MnO}_2 (\text{s}))} = 1.68 \text{ V}$; $E^0_{(\text{MnO}_2 (\text{s})/\text{Mn}^{2+} (\text{aq}))} = 1.21 \text{ V}$; $E^0_{(\text{Mn}^{2+} (\text{aq})/\text{Mn(s)})} = -1.03 \text{ V}$]

CEC085

9. The standard reduction potential data at 25°C is given below

$E^\circ (\text{Fe}^{3+}, \text{Fe}^{2+}) = +0.77 \text{ V}$;
$E^\circ (\text{Fe}^{2+}, \text{Fe}) = -0.44 \text{ V}$;
$E^\circ (\text{Cu}^{2+}, \text{Cu}) = +0.34 \text{ V}$;
$E^\circ (\text{Cu}^+, \text{Cu}) = +0.52 \text{ V}$;
$E^\circ [\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}] = +1.23 \text{ V}$;
$E^\circ [\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-] = +0.40 \text{ V}$;
$E^\circ (\text{Cr}^{3+}, \text{Cr}) = -0.74 \text{ V}$;
$E^\circ (\text{Cr}^{2+}, \text{Cr}) = -0.91 \text{ V}$;

Match E° of the redox pair in List-I with the values given in List-II and select the correct answer using the code given below the lists :

[JEE (Advanced) 2013]

List-I	List-II
(P) $E^\circ (\text{Fe}^{3+}, \text{Fe})$	(1) -0.18 V
(Q) $E^\circ (4\text{H}_2\text{O} \rightleftharpoons 4\text{H}^+ + 4\text{OH}^-)$	(2) -0.4 V
(R) $E^\circ (\text{Cu}^{2+} + \text{Cu} \rightarrow 2\text{Cu}^+)$	(3) -0.04 V
(S) $E^\circ (\text{Cr}^{3+}, \text{Cr}^{2+})$	(4) -0.83 V

- (A) P-4, Q-1, R-2, S-3 (B) P-2, Q-3, R-4, S-1
(C) P-1, Q-2, R-3, S-4 (D) P-3, Q-4, R-1, S-2

CEC071

10. Plotting $1/\Lambda_m$ against $c\Lambda_m$ for aqueous solutions of a monobasic weak acid (HX) resulted in a straight line with y-axis intercept of P and slope of S. The ratio P / S is
[Λ_m = molar conductivity]

Λ_m° = limiting molar conductivity

c = molar concentration

K_a = dissociation constant of HX]

[JEE (Advanced) 2023]

- (A) $K_a \Lambda_m^\circ$ (B) $K_a \Lambda_m^\circ / 2$
(C) $2 K_a \Lambda_m^\circ$ (D) $1 / (K_a \Lambda_m^\circ)$

CEC215

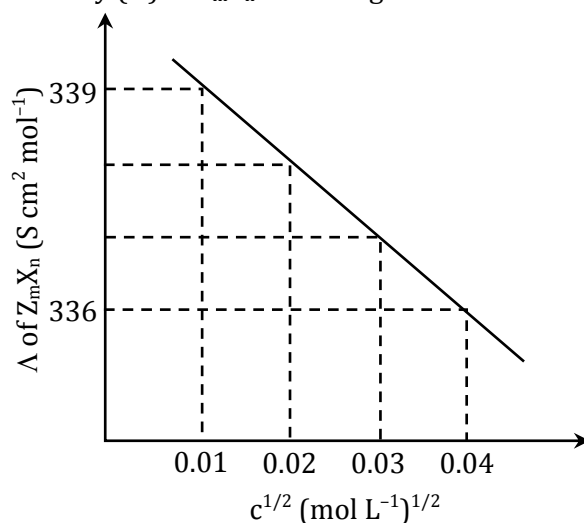
11. Consider the strong electrolytes Z_mX_n , U_mY_p and V_mX_n . Limiting molar conductivity (Λ°) of U_mY_p and V_mX_n are 250 and 440 $S\text{ cm}^2\text{ mol}^{-1}$, respectively. The value of (m + n + p) is _____.
Given :

Ion	$Z^{n\oplus}$	$U^{p\oplus}$	$V^{n\oplus}$	$X^{m\ominus}$	$Y^{m\ominus}$
$\lambda^\circ (S\text{ cm}^2\text{ mol}^{-1})$	50.0	25.0	100.0	80.0	100.0

λ° is the limiting molar conductivity of ions

The plot of molar conductivity (Λ) of Z_mX_n vs $c^{1/2}$ is given below.

[JEE (Advanced) 2022]



CEC086

Paragraph for Q.12 and Q.13

At 298 K, the limiting molar conductivity of a weak monobasic acid is $4 \times 10^2 S\text{ cm}^2\text{ mol}^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation of α and the molar conductivity is $y \times 10^2 S\text{ cm}^2\text{ mol}^{-1}$. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2 S\text{ cm}^2\text{ mol}^{-1}$.

12. The value of y is ____.

[JEE (Advanced) 2021]

CEC082

13. The value of α is ____.

[JEE (Advanced) 2021]

CEC083

14. The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 cm^2 . The conductance of this solution was found to be $5 \times 10^{-7} \text{ S}$. The pH of the solution is 4. The value of limiting molar conductivity (Λ_m^0) of this weak monobasic acid in aqueous solution is $Z \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$. The value of Z is.

[JEE (Advanced) 2017]

CEC077

15. The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda_{X^0}^0 \approx \lambda_{Y^0}^0$, the difference in their pK_a values, $pK_a(\text{HX}) - pK_a(\text{HY})$, is (consider degree of ionization of both acids to be $\ll 1$).

[JEE (Advanced) 2015]

CEC074

16. In a conductometric titration, small volume of titrant of higher concentration is added stepwise to a larger volume of titrate of much lower concentration, and the conductance is measured after each addition.

The limiting ionic conductivity (Λ_0) values (in $\text{mS m}^2 \text{ mol}^{-1}$) for different ions in aqueous solutions are given below :

Ions	Ag^+	K^+	Na^+	H^+	NO_3^-	Cl^-	SO_4^{2-}	OH^-	CH_3COO^-
Λ_0	6.2	7.4	5.0	35.0	7.2	7.6	16.0	19.9	4.1

For different combinations of titrates and titrants given in **List-I**, the graphs of 'conductance' versus 'volume of titrant' are given in **List-II**.

Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I

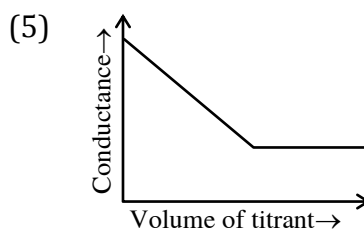
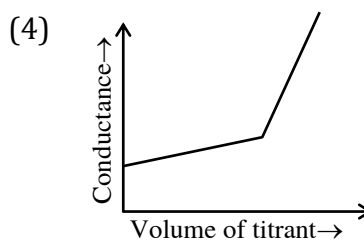
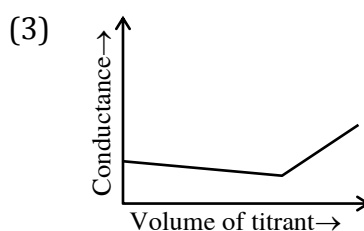
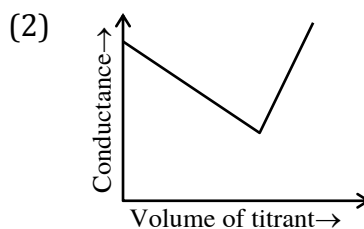
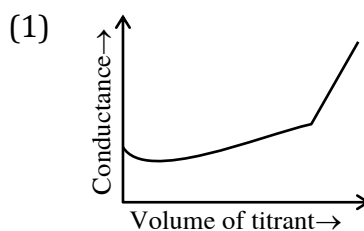
(P) Titrate : KCl
Titrant : AgNO_3

(Q) Titrate : AgNO_3
Titrant : KCl

(R) Titrate : NaOH
Titrant : HCl

(S) Titrate : NaOH
Titrant : CH_3COOH

List-II



(A) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 5$

(B) $P \rightarrow 2, Q \rightarrow 4, R \rightarrow 3, S \rightarrow 1$

(C) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 2, S \rightarrow 5$

(D) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$

[JEE (Advanced) 2024]

CEC287

17. An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : [JEE (Advanced) 2013]

List-I		List-II
(P)	$(\text{C}_2\text{H}_5)_3\text{N} + \text{CH}_3\text{COOH}$ X Y	(1) Conductivity decreases and then increases
(Q)	$\text{KI}(0.1\text{M}) + \text{AgNO}_3(0.01\text{M})$ X Y	(2) Conductivity decreases and then does not change much
(R)	$\text{CH}_3\text{COOH} + \text{KOH}$ X Y	(3) Conductivity increases and then does not change much
(S)	$\text{NaOH} + \text{HI}$ X Y	(4) Conductivity does not change much and then increases

(A) P-3, Q-4, R-2, S-1 (B) P-4, Q-3, R-2, S-1 (C) P-2, Q-3, R-4, S-1 (D) P-1, Q-4, R-3, S-2

CEC072

18. An aqueous solution of hydrazine (N_2H_4) is electrochemically oxidized by O_2 , thereby releasing chemical energy in the form of electrical energy. One of the products generated from the electrochemical reaction is $\text{N}_2(\text{g})$.

Choose the correct statement(s) about the above process :

- (A) OH^- ions react with N_2H_4 at the anode to form $\text{N}_2(\text{g})$ and water, releasing 4 electrons to the anode.
 (B) At the cathode, N_2H_4 breaks to $\text{N}_2(\text{g})$ and nascent hydrogen released at the electrode reacts with oxygen to form water.
 (C) At the cathode, molecular oxygen gets converted to OH^- .
 (D) Oxides of nitrogen are major by-products of the electrochemical process.

[JEE (Advanced) 2024]

CEC288

19. All the energy released from the reaction $\text{X} \rightarrow \text{Y}$, $\Delta_r G^\circ = -193 \text{ kJ mol}^{-1}$ is used for the oxidizing $\text{M}^\oplus$ and $\text{M}^\oplus \rightarrow \text{M}^{3\oplus} + 2\text{e}^\ominus$, $E^\circ = -0.25 \text{ V}$.

Under standard conditions, the number of moles of $\text{M}^\oplus$ oxidized when one mole of X is converted to Y is : [$F = 96500 \text{ C mol}^{-1}$]

[JEE (Advanced) 2015]

CEC075

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	B	A	A	B	D	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	C	B	C	C	C	A	B	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	D	C	D	C	A	B	A	C
Que.	31	32	33	34	35	36	37	38	39	
Ans.	C	A	D	B	B	B	D	D	B	

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	AD	C	D	A	D	A	C	C	A

EXERCISE (S)

- | | | | | |
|--------------------------------|------------------------------------|--------------------|-----------|----------|
| 1. 2 | 2. 3 | 3. 26 | 4. 3 | 5. 2500 |
| 6. 125 | 7. 9 | 8. 10 | 9. 150 | 10. 400 |
| 11. 60 | 12. 4 | 13. 2 | 14. 0.78 | |
| 15. 0.53, 0.38 | 16. 0.80 | 17. 0.05 or 0.06 | 18. 0.41 | 19. 3.86 |
| 20. 7.60 | 21. 0.59 | 22. 250.00, 125.00 | 23. -0.16 | 24. 9.40 |
| 25. (a) 0.10 (b) 0.03 (c) 0.30 | 26. (a) 0.75 ; (b) 0.69 ; (c) 1.10 | | | |

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	1	2	3	3	2	4	18	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	275	3	917	1	2	1	3	338 OR 339	17	6
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	150	2	4	25	3	4	1	2	3
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	25	66	2	2	3	4	4	3	2
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	107 gm or 108	45.75 Or 46	11	5	161	3	4	1	2	1
Que.	51	52								
Ans.	4	3								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AB	13.32	B	10	D	ABC	-11.62	0.77	D	A
Que.	11	12	13	14	15	16	17	18	19	
Ans.	7	0.86	0.21 or 0.22	6	3	C	A	AC	4	